

S. No.	Component	Specification	Purpose
1	2 × BLDC Motor + 2 × ESC	150A, 20A ESC, 3670 BLDC	Forward/Backward and Left/Right Motion
2	Stepper Motor + Driver	NEMA Stepper + A4988	Depth control via syringe lead screw mechanism
3	STM32F4 Blackpill	Cortex-M4 MCU	Motor and LED control
4	Raspberry Pi 5	Quad-core 2.4GHz, 8GB RAM	Server hosting, camera and ML integration
5	Limit Switches	Mechanical	Restrict syringe travel
6	LEDs	High-brightness LEDs	Illumination in turbid water
7	Li-ion/LiPo Battery	12V DC	Powering all modules
8	Battery Management System (BMS)	40A, 12.6V BMS	Overcharge and over-discharge protection
9	Sony IMX462 Arducam	Ultra Low-Light Camera	Underwater image/video capture

Component	Specification	Description
12v Battery Management sys ^m (BMS) + 3 Batteries (series)	BMS: 3S, 40A, 12.6V Batteries: 3X 18650 Lithium Ion	Protection from overcharge / discharge Short ckt, cell Balancing
• Stepper Motor Driver	A4988 board	• takes controls from STM • sends to stepper • powers the motor
• LED driver Module	JZ-LEDPower (Ip: 7v-30Vdc)	• Constant current o/p for LED • PWM pin for Dimming
• STM32 Black Pill	• STM32F4 (Black Pill) ↳ ARM Cortex-M4 32bit RISC core	
• 4L-99 Impact Switch	• 4L-99 Impact Switch 3Pin (GND, OUT, VCC), 2A / 125VAC	• fan collision detection
• RS2205 Brushless DC Motor	• EMAX RS2205 2300KV BLDC • CW (Clockwise) • 12.6-16.8V	
• FLYCOLOR 150A waterproof Brushless ESC	• 150A / 7.4-22.2V (2-6S LiPo Batt) • Waterproof	
• Cymon 20A BLHeli-S DSHOT OPTO ESC	• 20A continuous current (P/P) • BLHeli-S firmware (2S-6S LiB)	
• Sunpass Hobby Rocket-RC 3670 1200KV Brushless Inrunner Motor	• 4 Pole Brushless Inrunner Motor • 12000KV (4S-6S Lipo)	

